

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART I

THEORY

March 7, 1981

Instructions:

1. Open Book and Open Notes
2. Do four out of the following six problems
3. Use a separate blue book for this section of the exam
4. Indicate which questions you wish graded

1. Let X_1, X_2, X_3 be i.i.d. random variates with common density

$$f(x) = e^{-x}, \quad 0 < x < \infty$$

Let $Y_1 = X_1 + X_2$; $Y_2 = X_2 + X_3$.

- a) Find $\text{Var}(Y_1 + Y_2)$
- b) Find correlation between Y_1 and Y_2 .

2. Let X be a discrete random variate with

$$\Pr(X = x) = p^{x-1} - p^x \quad \text{for } x = 1, 2, 3, \dots \quad ; p \text{ a constant, } 0 \leq p \leq 1.$$

We observe $X = 2$. Give a 90% confidence interval for p of the form $p < c$.
(That is, find c .)

3. Let Y be distributed with density

$$f(y) = 1 \quad \text{for } 1 \leq y \leq 2.$$

Let $X = \log_e (Y^2 + 6Y + 9)$

find the moment generating function of X .

4. Let X be a discrete random variate with

$$\Pr(X = x) = e^{-\alpha} \alpha^x / x! \quad , x = 0, 1, 2, \dots \quad ; \alpha > 0.$$

Given that $E(X^2) = 7 E(X)$, find α .

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART II

APPLIED

March 7, 1981

- Instructions:
1. Open Book and Open Notes
 2. Do all three problems
 3. Use a separate blue book for this section of the exam

1. For a population of students an educational psychologist recorded for each student their IQ and their performance on a spatial task. The data on the population (of 10,000 students) gave the following results:

Mean IQ = 100

Standard deviation IQ = 15

Mean on spatial performance 70

Standard deviation on spatial performance 5

The correlation between IQ and spatial performance was 0.6.

- a) For students with an IQ of 130, what is the expected performance on the spatial task?
 - b) (Assuming that performance on the spatial task for a given IQ is normally distributed) what percent of students with IQ's of 130 will score 70 or higher on the spatial performance task?
-

2. An industrial engineer sought to determine whether color of lighting affected productivity of assembly and inspection workers. Four assembly workers were randomly assigned: two to cool light and two to warm light. Four inspection workers were similarly randomly assigned. Each worker's performance was graded on a scale (approximately normal) from 0 to 100 (where 100 is excellent performance). The scores were as follows:

<u>Worker</u>	<u>Type of Work</u>	<u>Type of lighting</u>	<u>Performance Score</u>
Jones	assembly	cool	65
Smith	assembly	cool	67
Green	assembly	warm	76
Black	assembly	warm	80
Doe	inspection	cool	74
Davis	inspection	cool	78
Marks	inspection	warm	79
Stone	inspection	warm	81

Carry out appropriate tests (at the 0.05 level of significance) of whether lighting makes a difference in performance and summarize relevant conclusions.

3. A social scientist sought to study the relation between the type of upbringing (lenient vs. strict) that parents applied to their children, and the type of upbringing that the parents experienced when they were children. 100 parents (all mothers) were studied, and the following results were found. Of the 100 parents, 60 were brought up in strict households, 40 were brought up in lenient households. Of these same 100 parents, 50 brought up their children strictly, and 50 brought up their children leniently. Of the 60 parents who were brought up strictly, 36 brought up their children strictly. Given these facts test (at the 0.05 level) whether the upbringing of the parents were independent of how they brought up their children.

March 7, 1981

5. Let $X_1, X_2, \dots, X_n$ be independently and identically distributed random variates each with density $f(x)$. Test:

Null Hypothesis: $f(x) = 1$, for $0 \leq x \leq 1$,
 $= 0$, otherwise.

against the

Alternative Hypothesis: $f(x) = 0.24$ for $0 \leq x < 0.3$
 $= 0.40$ for $0.3 \leq x < 1$,
 $= 0$, otherwise.

(show test statistic and region of rejection of test).

6. Let $X_1, \dots, X_n$ be independently and identically distributed random variates each having density function

$$g(x, \theta) = x^\theta e^{-x} / \theta! \quad \text{if } x > 0 \\ = 0 \quad \text{otherwise,}$$

for θ a positive integer.

Use the Neyman-Pearson Lemma to determine the nature of the best critical region for testing : Null Hypothesis: $\theta = 3$
against: Alternative Hypothesis: $\theta = 4$.

Rutgers-The State University
Master's Comprehensive Examination

Part II

Applied

December 5, 1981

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
-

1. Galton investigated 78 families, classifying children according to whether or not they are light eyed, whether or not they had a light eyed parent, and whether or not they had a light eyed grand parent. The following 2X2X2 table resulted:

		Grandparent			
		Light		Not	
		Parent			
		Light	Not	Light	Not
Child	Light	1928	552	596	508
	Not	303	395	225	501

Test for complete independence at .01 level.

2. Steel brackets were sent to three different vendors to be zinc plated: 12 brackets were plated, four by each vendor. The thickness of plating (in 10^{-5} inches) are recorded in the table below:

	Vendor		
	A	B	C
Zinc Plating (in 10^{-5} inches)	40	25	27
	38	32	24
	30	13	20
	47	35	13

A question of interest is whether there was a difference in thickness among the vendors. Construct an ANOVA table for no difference.

3. Raw material used in the production of a synthetic fiber is stored in a place which has no humidity control. Measurements of the relative humidity in the storage place and the moisture content of a sample of the raw material (both in percentage) on 12 days yielded the following result:

x	y
<u>Humidity</u>	<u>Moisture Content</u>
42	12
35	8
50	14
43	9
48	11
62	16
31	7
36	9
44	12
39	10
55	13
48	11

- (a) Estimate α and β , the coefficients of regression line of y on x .
 - (b) Find a 0.99 confidence interval for the mean moisture content of the raw material when the humidity of the storage place is 40%
 - (c) Find 0.95 limits of prediction for the moisture content of the raw material when the humidity of storage place is 40%
 - (d) Referring to (c), indicate to what extent the width of the interval is affected by the size of the sample and to what extent it is affected by the inherent variability of the data.
4. Comparing two kinds of emergency flares, a consumer testing service obtained the following burning times (rounded to the nearest 10th of a minute):
- Brand C: 19.4, 21.5, 15.3, 17.4, 16.8, 16.6, 20.3, 22.5, 21.3, 23.4, 19.7, 21.0.
- Brand D: 16.5, 15.8, 24.7, 10.2, 13.5, 15.9, 15.7, 14.0, 12.1, 17.4, 15.6, 15.8.

Use a nonparametric test and a level of significance of 0.01 to check whether it is reasonable to say that there is no difference between the true average burning times of the two kinds of flares. State the model for the validity of the test you use.

5. A test of four detergents (I, II, III, IV) using four methods of application at four hospitals was conducted so that each detergent was used exactly once with each method and exactly once in each hospital. The assignment of detergents to each method-hospital combination and the observed data together with some preliminary calculations are shown in the following table.

6. A new variety of tomato has been developed. It is desired to know if the yield of this variety is greater than the yield of a standard variety. An experiment was conducted and the yield was observed. The results are summarized below.

<u>New Variety</u>	<u>Standard Variety</u>
$n_N = 15$	$n_S = 10$
$\bar{x}_N = 125$	$\bar{x}_S = 100$
$s_N^2 = 85.7$	$s_S^2 = 94.4$

Making the usual assumptions (including equality of variances) answer the following questions:

- One person says the t-test is the appropriate procedure. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Another person says that the Analysis of Variance definitely should be used. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Do the results in parts (a) and (b) agree? If yes, explain why. If no, which procedure is correct and explain why.

	1	2	3	4	$\bar{Y}_{i..}$
1	8.7 (I)	9.2 (II)	11.6 (III)	9.1 (IV)	9.650
2	7.5 (II)	12.7 (III)	4.6 (IV)	7.3 (I)	8.025
3	14.0 (III)	9.2 (IV)	5.1 (I)	6.7 (II)	8.750
4	11.3 (IV)	8.7 (I)	4.0 (II)	12.9 (III)	9.225
$\bar{Y}_{.j.}$	10.375	9.950	6.325	9.000	$\bar{Y} = 8.9125$

- (a) Is this a complete block design?
- (b) What are the blocking factors?
- (c) Complete the following ANOVA table.

Analysis of Variance Table for Detergent

Data				
Source of Variation	d.f.	SS	M.S.	V.R.
Methods				
Hospitals				
Detergents				
Residual Error				
Total		136.097		

- (d) Test whether there are differences in the mean response due to the detergents.
- (e) Test whether there are differences in the mean response due to the hospitals.
- (f) Test whether there are differences in the mean response due to the methods of application.

6. A new variety of tomato has been developed. It is desired to know if the yield of this variety is greater than the yield of a standard variety. An experiment was conducted and the yield was observed. The results are summarized below.

<u>New Variety</u>	<u>Standard Variety</u>
$n_N = 15$	$n_S = 10$
$\bar{x}_N = 125$	$\bar{x}_S = 100$
$s_N^2 = 85.7$	$s_S^2 = 94.4$

Making the usual assumptions (including equality of variances) answer the following questions:

- One person says the t-test is the appropriate procedure. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Another person says that the Analysis of Variance definitely should be used. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Do the results in parts (a) and (b) agree? If yes, explain why. If no, which procedure is correct and explain why.

Rutgers-The State University
Master's Comprehensive Examination

December 5, 1981

Theory

Part I

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

-
1. A major car manufacturer has four plants in operation, located in Flint, Pontiac, Detroit and Mahwah. The Flint and Pontiac plants produce about 30% of all cars each, while the Detroit and Mahwah plants produce 20% each. It is known that 5% of Flint cars are "lemons" and so are 4% of Pontiac cars, and 2% of Detroit and Mahwah cars. If you were one of the lucky people who bought a "lemon" from this company, what is the probability that it came from the Pontiac plant (assume that your dealer's lot has cars coming from each of the four plants in the same proportions as the contribution of each plant to the total production volume of the manufacturer).
 2. Let X, Y be independent $N(0,1)$ random variables. Find the density for $R = \sqrt{X^2 + Y^2}$.
 3. Three light bulbs are picked at random from a lot. Assuming that the distribution of the life time for a single bulb is exponential with mean 2 years, what is the probability that the average life time of the three bulbs chosen will be more than 1.9 years?
 4. Let $X_1, \dots, X_n$ be a random sample from a normal distribution with known mean μ and variance σ^2 unknown. Is $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ an efficient estimator of σ^2 ? If not, what is an efficient estimator of σ^2 ? (Recall that an unbiased estimator is efficient iff its variance attains the Cramer-Rao lower bound.)
 5. Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with density:

$$f(x, \theta) = \begin{cases} \frac{2x}{\theta^2} & 0 < x < \theta \\ \theta & \\ 0 & \text{otherwise} \end{cases}, \text{ where } \theta > 0.$$

- (i) Find a maximum likelihood estimator of θ .
- (ii) What is a sufficient statistic for θ ?
6. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean λ .
 - (i) Find the uniformly most powerful test of $H_0: \lambda = \lambda_0$ versus $H_a: \lambda > \lambda_0$.
 - (ii) Sketch the power function of this test for $\lambda_0 = 1$ and $n=25$. (Take $\alpha = .05$ and use the central limit theorem approximation.)

Rutgers-The State University
Master's Comprehensive Examination

Part II

Applied

December 5, 1981

- Instructions:
1. Open book and open notes
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
-

1. Galton investigated 78 families, classifying children according to whether or not they are light eyed, whether or not they had a light eyed parent, and whether or not they had a light eyed grand parent. The following 2x2x2 table resulted:

		Grandparent			
		Light		Not	
		Parent			
		Light	Not	Light	Not
Child	Light	1928	552	596	508
	Not	303	395	225	501

Test for complete independence at .01 level.

2. Steel brackets were sent to three different vendors to be zinc plated: 12 brackets were plated, four by each vendor. The thickness of plating (in 10^{-5} inches) are recorded in the table below:

	Vendor		
	A	B	C
Zinc Plating (in 10^{-5} inches)	40	25	27
	38	32	24
	30	13	20
	47	35	13

A question of interest is whether there was a difference in thickness among the vendors. Construct an ANOVA table for no difference.

3. Raw material used in the production of a synthetic fiber is stored in a place which has no humidity control. Measurements of the relative humidity in the storage place and the moisture content of a sample of the raw material (both in percentage) on 12 days yielded the following result:

x	y
<u>Humidity</u>	<u>Moisture Content</u>
42	12
35	8
50	14
43	9
48	11
62	16
31	7
36	9
44	12
39	10
55	13
48	11

- (a) Estimate α and β , the coefficients of regression line of y on x .
 - (b) Find a 0.99 confidence interval for the mean moisture content of the raw material when the humidity of the storage place is 40%.
 - (c) Find 0.95 limits of prediction for the moisture content of the raw material when the humidity of storage place is 40%.
 - (d) Referring to (c), indicate to what extent the width of the interval is affected by the size of the sample and to what extent it is affected by the inherent variability of the data.
4. Comparing two kinds of emergency flares, a consumer testing service obtained the following burning times (rounded to the nearest 10th of a minute):
- Brand C: 19.4, 21.5, 15.3, 17.4, 16.8, 16.6, 20.3, 22.5, 21.3, 23.4, 19.7, 21.0.
- Brand D: 16.5, 15.8, 24.7, 10.2, 13.5, 15.9, 15.7, 14.0, 12.1, 17.4, 15.6, 15.8.
- Use a nonparametric test and a level of significance of 0.01 to check whether it is reasonable to say that there is no difference between the true average burning times of the two kinds of flares. State the model for the validity of the test you use.
5. A test of four detergents (I, II, III, IV) using four methods of application at four hospitals was conducted so that each detergent was used exactly once with each method and exactly once in each hospital. The assignment of detergents to each method-hospital combination and the observed data together with some preliminary calculations are shown in the following table.

	1	2	3	4	$\bar{Y}_{i..}$
1	8.7 (I)	9.2 (II)	11.6 (III)	9.1 (IV)	9.650
2	7.5 (II)	12.7 (III)	4.6 (IV)	7.3 (I)	8.025
3	14.0 (III)	9.2 (IV)	5.1 (I)	6.7 (II)	8.750
4	11.3 (IV)	8.7 (I)	4.0 (II)	12.9 (III)	9.225
$\bar{Y}_{.j.}$	10.375	9.950	6.325	9.000	$\bar{Y} = 8.9125$

- (a) Is this a complete block design?
 (b) What are the blocking factors?
 (c) Complete the following ANOVA table.

Analysis of Variance Table for Detergent

Data				
Source of Variation	d.f.	SS	M.S.	V.R.
Methods				
Hospitals				
Detergents				
Residual Error				
Total		136.097		

- (d) Test whether there are differences in the mean response due to the detergents.
 (e) Test whether there are differences in the mean response due to the hospitals.
 (f) Test whether there are differences in the mean response due to the methods of application.

6. A new variety of tomato has been developed. It is desired to know if the yield of this variety is greater than the yield of a standard variety. An experiment was conducted and the yield was observed. The results are summarized below.

<u>New Variety</u>	<u>Standard Variety</u>
$n_N = 15$	$n_S = 10$
$\bar{x}_N = 125$	$\bar{x}_S = 100$
$s_N^2 = 85.7$	$s_S^2 = 94.4$

Making the usual assumptions (including equality of variances) answer the following questions:

- One person says the t-test is the appropriate procedure. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Another person says that the Analysis of Variance definitely should be used. Assuming this person is correct complete the analysis of these data including the construction of the null and alternative hypotheses.
- Do the results in parts (a) and (b) agree? If yes, explain why. If no, which procedure is correct and explain why.